Radix Sort

$$T(n) = \Theta\left(\frac{b}{r}\left(n + 2^r\right)\right) = I$$

Case 2: $b \geq \log_2 n$ (Finding good choice for r)

then $r = \log_2 n$.

Linear if $\theta\left(\frac{b}{\log_2 n}\right) \in \theta(1)$.

Case 1: $b < \log_2 n$.

$(r \leq b)$ known.

$$I = \Theta\left(\frac{b}{r}(n + 2^r)\right)$$

Take $r = b$,

$$I = \Theta\left(\frac{b}{b}(n + 2^b)\right)$$

$$= \Theta(n + n)$$

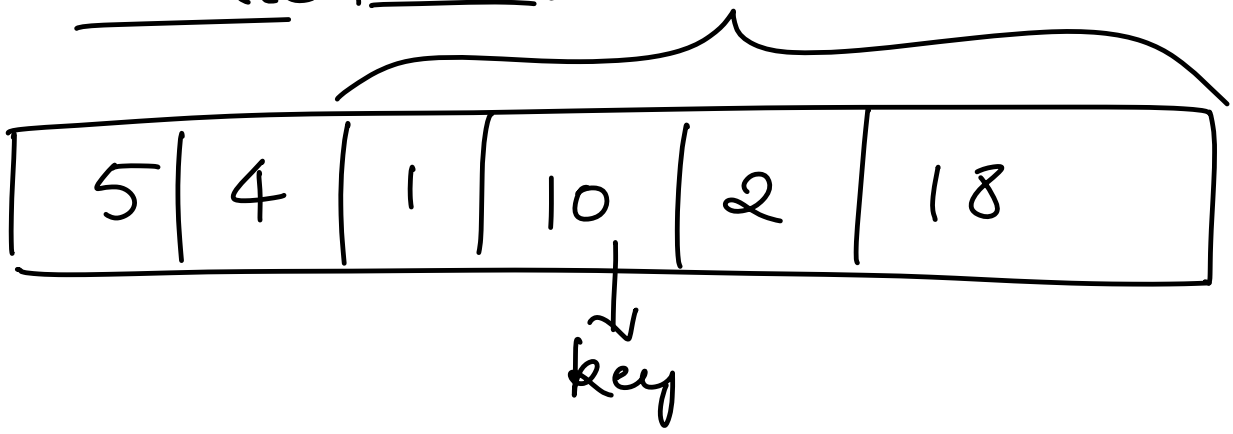
$$= \Theta(n) \quad \text{(Linear)}$$

$b < \log_2 n$

$2^b < 2^{\log_2 n}$

$2^b < n$

Insertion Sort:



Card in hand

In hand 5 $j=2$

In hand 1 4 5 $j=3$

In hand 1 4 5 10

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In hand 1 2 4 5 10

In hand 1 2 4 5 10 18

$A[1, \ldots, j] \rightarrow$ In hand sorted

Key $\rightarrow A[j]$

$A[j+1, \ldots, n] \rightarrow$ Cards in table

Insertion Sort:

1. For $j \leftarrow 2$ to n
2. $\quad key \leftarrow A[j]$
3. $\quad i \leftarrow j-1$
4. $\quad$ while $i>0$ and $A[i] > key$
5. $\quad\quad A[i+1] \leftarrow A[i]$
6. $\quad\quad i \leftarrow i-1$
7. $\quad A[i+1] \leftarrow key$

Worst Case:

$$T(n) = \sum_{j=2}^{n} \sum_{i=j-1}^{n-1} 1$$

$$= \sum_{j=2}^{n} (n-1-(j-1)+1)$$

$$= \sum_{j=2}^{n} n-j+1$$

$$= (n-1)+(n-2)+\cdots 1$$

$$= (n-1)n/2 \in \Theta(n^2).$$

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Any number

n → even → $n/2$

n → odd → $3n+1$

"3n+1 Conjecture"

or

"Collad's Conjecture".

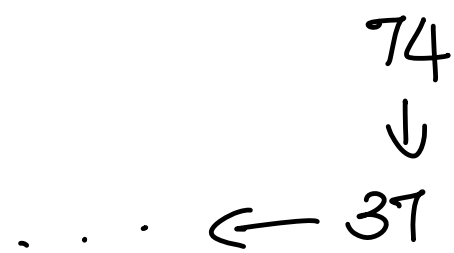
$13 \to 40 \to 20 \to 10 \to 5 \to 16 \to 8 \to 4 \to 2 \to 1$

$3(13)+1$

$5(3)+1$

$19 \xrightarrow{3(9)+1} 58 \longrightarrow 29 \xrightarrow{3(29)+1} 87 \xrightarrow{3(87)+1} 262$

$262 \downarrow 131 \downarrow 394 \downarrow 197 \downarrow 592 \downarrow 296 \downarrow 148 \downarrow$


74
37

Bucket Sort:

Assumption: The elements are uniformly distributed over $[0,1)$.

Bucket-Sort (A):

1. $n \leftarrow$ length(A)
2. For $j \leftarrow 1$ to n
3. do insert $A[j]$ into auxiliary array B of linked list $B[\lfloor nA[j] \rfloor]$
4. For $i \leftarrow 0$ to $n-1$
5. do Sort list $B[i]$ with insertion sort
6. Concatenate the lists $B[0], B[1], \ldots B[n-1]$ in order.

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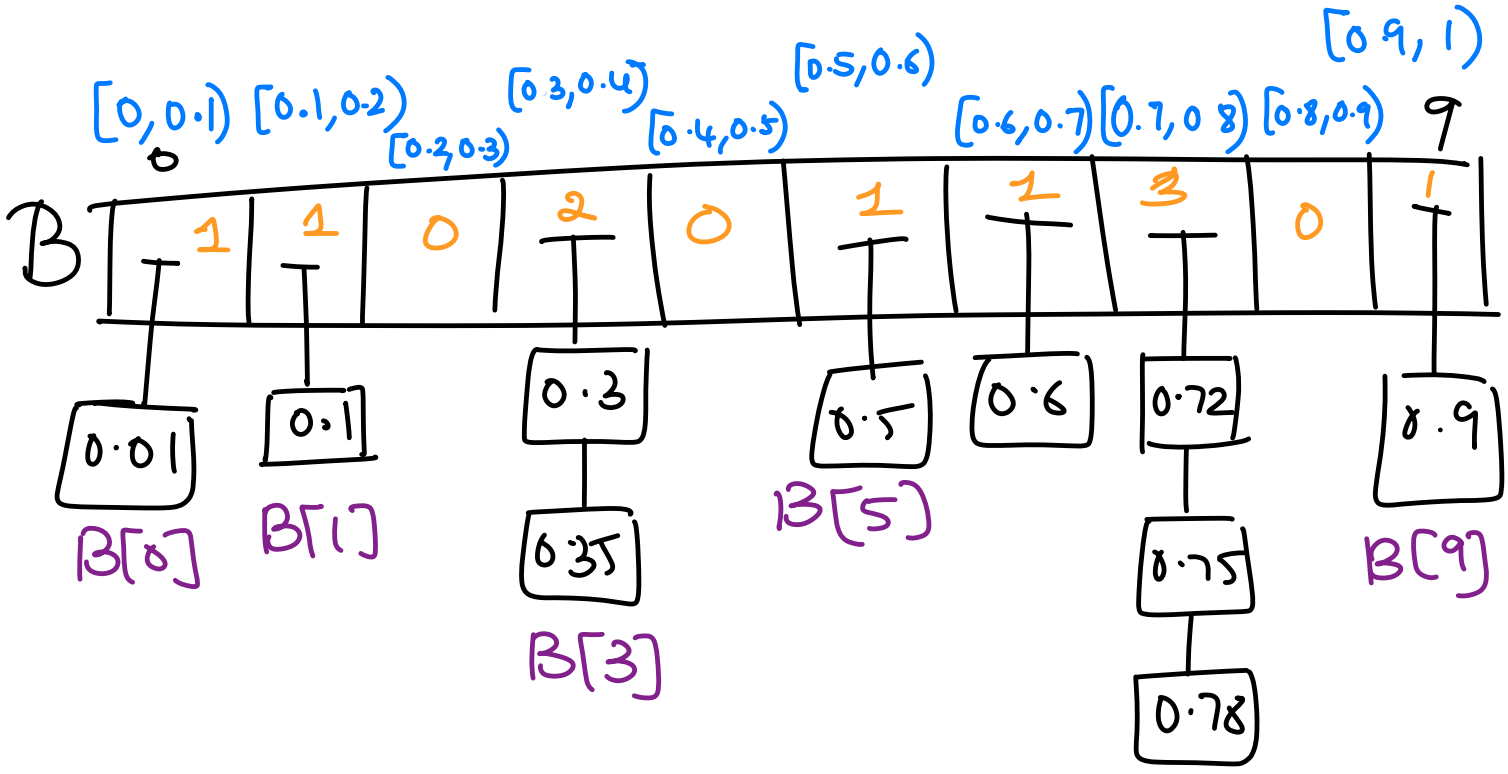
Worst Case Analysis:

Since worst case time complexity of Insertion Sort is $\Theta(n^2)$, Bucket sort is also $\Theta(n^2)$.

eg:

10

A | 0.1 | 0.72 | 0.5 | 0.3 | 0.01 | 0.75 | 0.35 | 0.78 | 0.6 | 0.9 |



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• → no of elts in a bucket.

$n \leftarrow 10.$

$i = 1.$

$A[1] = 0.1$

Uniform Distribution

$$h = \frac{b-a}{n}$$

$$= \frac{1-0}{10}$$

$$= \frac{1}{10}$$

$$= 0.1.$$

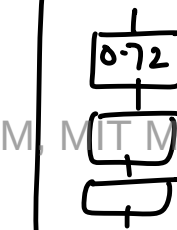
Concatenation

0.01, 0.1, 0.3, 0.35, 0.5, 0.6, 0.72, 0.75, 0.78, 0.9

Concatenation is sorted answer.

Note: Uniform distribution is required, else we end up doing worst case of insertion sort.

eg: [0.7, 0.8)



* Worst Case is all n numbers falling inside one bucket which gives $\theta(n^2)$ time due to insertion sort.

Let $X_{ij} = \begin{cases} 1 & \text{if } A[j] \text{ falls inside in bucket } B[i] \\ 0 & \text{otherwise} \end{cases}$

↑ Indicates whether a no. falls inside a bucket or not.

$$\sum_{j=1}^{n} X_{ij} = \text{No of elements in bucket } B[i].$$

$$= n_i.$$

Each bucket B[i] takes $\Theta(n_i^2)$ time (insertion sort)

Hence, time complexity of Bucket Sort

$$T(n) = \sum_{i=0}^{n-1} \Theta(n_i^2) + \Theta(n)$$

↑ Concatenation

Average Case Analysis:

$$E[T(n)] = \sum_{i=0}^{n-1} O\left(E(n_i^2)\right) + \Theta(n).$$

$$E(n_i^2) = E\left(\left(\sum_{j=1}^{n} X_{ij}\right)^2\right)$$

$$= E\left(\sum_{j=1}^{n} X_{ij}^2 + \sum_{k=1}^{n} \sum_{\substack{j=1 \\ j \neq k}}^{n} X_{ij} X_{ik}\right)$$

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$$(X_{i1} + X_{i2})^2 = X_{i1}^2 + X_{i2}^2 + X_{i1}X_{i2} + X_{i2}X_{i1}$$

$$((X_{i1} + X_{i2}) + X_{i3})^2$$

$$= X_{i1}^2 + X_{i2}^2 + X_{i3}^2 + X_{i1}X_{i2} + X_{i2}X_{i1} + X_{i1}X_{i3} + X_{i2}X_{i3} + X_{i3}X_{i1} + X_{i3}X_{i2}$$

$$= X_{i1}^2 + X_{i2}^2 + X_{i3}^2 + \sum_{\substack{k=1 \\ j \neq k}}^{3} \sum_{j=1}^{3} X_{ij}X_{ik}$$

$$E(X_{ij}) = 1 \cdot P\left(\begin{matrix}A[j] \text{ falls} \\ \text{in } B[i]\end{matrix}\right) + 0 \cdot P\left(\begin{matrix}\text{does not} \\ \text{fall}\end{matrix}\right)$$

$$= 1 \cdot \frac{1}{n} + 0 \cdot \left(1 - \frac{1}{n}\right)$$

$$= \frac{1}{n}.$$

$$E(X_{ij}^2) = 1^2 \cdot P\left(\begin{matrix}A[j] \\ \text{falls in} \\ B[i]\end{matrix}\right) + 0^2 \cdot P\left(\begin{matrix}\text{does not} \\ \text{fall}\end{matrix}\right)$$

$$= \frac{1}{n}.$$